The BLEU Score

- The most common measure is BLEU (Papineni et al., 2002)
- It determines number of *n*-grams of various sizes that the MT output shares with the reference translations
- Computes a modified precision measure of the *n*-grams in MT result
- Since precision-based measures favor short sentences, a penalty term is introduced for long translations

The BLEU Score

Cand 1: Mary no slap the witch green.
Cand 2: Mary did not give a smack to a green witch.

Ref 1: Mary did not slap the green witch.
Ref 2: Mary did not smack the green witch.
Ref 3: Mary did not hit a green sorceress.

Cand 1 Unigram Precision: 5/6

BLEU Score

Cand 1: Mary no slap the witch green.
Cand 2: Mary did not give a smack to the green witch.

Ref 1: Mary did not slap the green witch.
Ref 2: Mary did not smack the green witch.
Ref 3: Mary did not hit a green sorceress.

Cand 1 Bigram Precision: 1/5

BLEU Score

Cand 1: Mary no slap the witch green.
Cand 2: Mary did not give a smack to the green witch.

Ref 1: Mary did not slap the green witch.
Ref 2: Mary did not smack the green witch.
Ref 3: Mary did not hit a green sorceress.

Cand 2 Unigram Precision: 7/10

BLEU Score

Cand 1: Mary no slap the witch green.
Cand 2: Mary did not give a smack to the green witch.

Ref 1: Mary did not slap the green witch.
Ref 2: Mary did not smack the green witch.
Ref 3: Mary did not hit a green sorceress.

Cand 2 Bigram Precision: 4/9

Modified N-Gram Precision

- If we evaluate performance on a single candidate *c* against a set of references, then for each n-value (n=1,…,N):

$$p_n = \frac{\text{number of n-grams in } c \text{ that appear in at least one of the references}^*}{\text{number of n-grams in } c}$$

- We compute the geometric mean of *n*-gram precision over all *n*-grams up to size *N* (typically 4):

$$p = \left(\prod_{n=1}^{N} p_n \right)^{\frac{1}{N}}$$

* - if an n-gram appears multiple times in *c*, we do not count it more than the maximal number of times it appears in any of the references

Modified *N*-Gram Precision

- BLEU can be easily extended to evaluate many sets of candidates and references (a corpus)
  - We do not cover it here

- Returning to the example from before we get: (for N=2)

Cand 1: $p = \sqrt[2]{\frac{5}{6}\frac{1}{5}} = 0.408$

Cand 2: $p = \sqrt[2]{\frac{7}{10}\frac{4}{9}} = 0.558$

Brevity Penalty

- Not easy to compute recall to complement precision since there are multiple alternative gold-standard references and don’t need to match all of them
- Instead, use a penalty for translations that are shorter than the reference translations
- Define effective reference length, *r*, for each sentence as the length of the reference sentence with the largest number of *n*-gram matches. Let *c* be the candidate sentence length:

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

BLEU Score

- Final BLEU Score: BLEU = $BP \times p$

Cand 1: Mary no slap the witch green.

Best Ref: Mary did not slap the green witch.

$$c = 6, \quad r = 7, \qquad BP = e^{(1-7/6)} = 0.846$$

$$BLEU = 0.846 \times 0.408 = 0.345$$

Cand 2: Mary did not give a smack to a green witch.

Best Ref: Mary did not smack the green witch.

$$c = 10, \quad r = 7, \qquad BP = 1$$

$$BLEU = 1 \times 0.558 = 0.558$$

The BLEU Score

- BLEU has been shown to correlate with human evaluation when comparing outputs from different SMT systems
- However, it is does not correlate with human judgments when comparing SMT systems with human translations
- Other MT evaluation metrics have been proposed that claim to overcome some of the limitations of BLEU